🌲 DAY 4: ARROW WORKFLOWS & TOPIC TREES

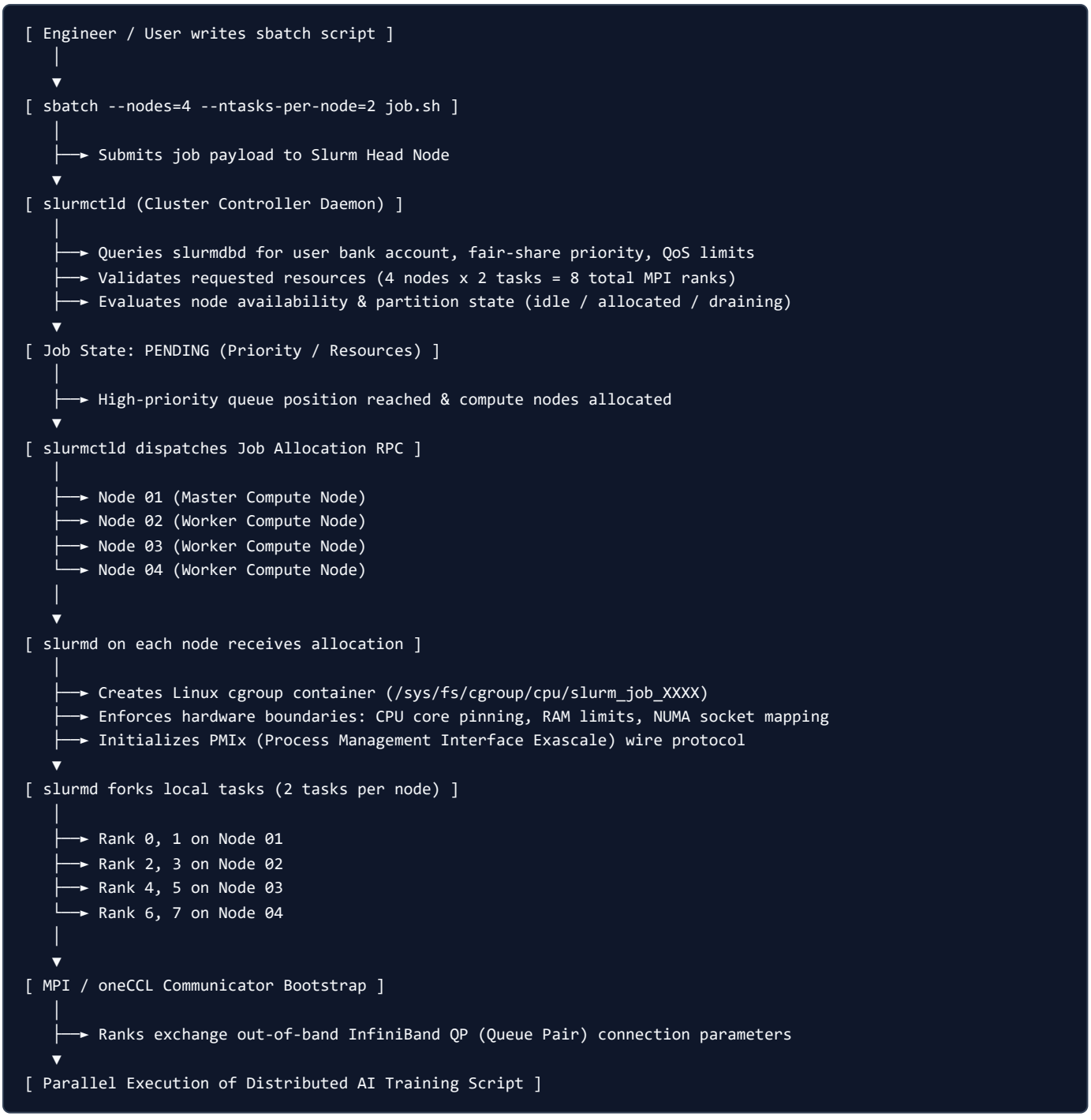
Distributed AI Training, Slurm Multi-Node Clusters & Intel VTune Profiling Optimization

Visual Step-by-Step Execution Pathways for HPC Engineers & Trainers

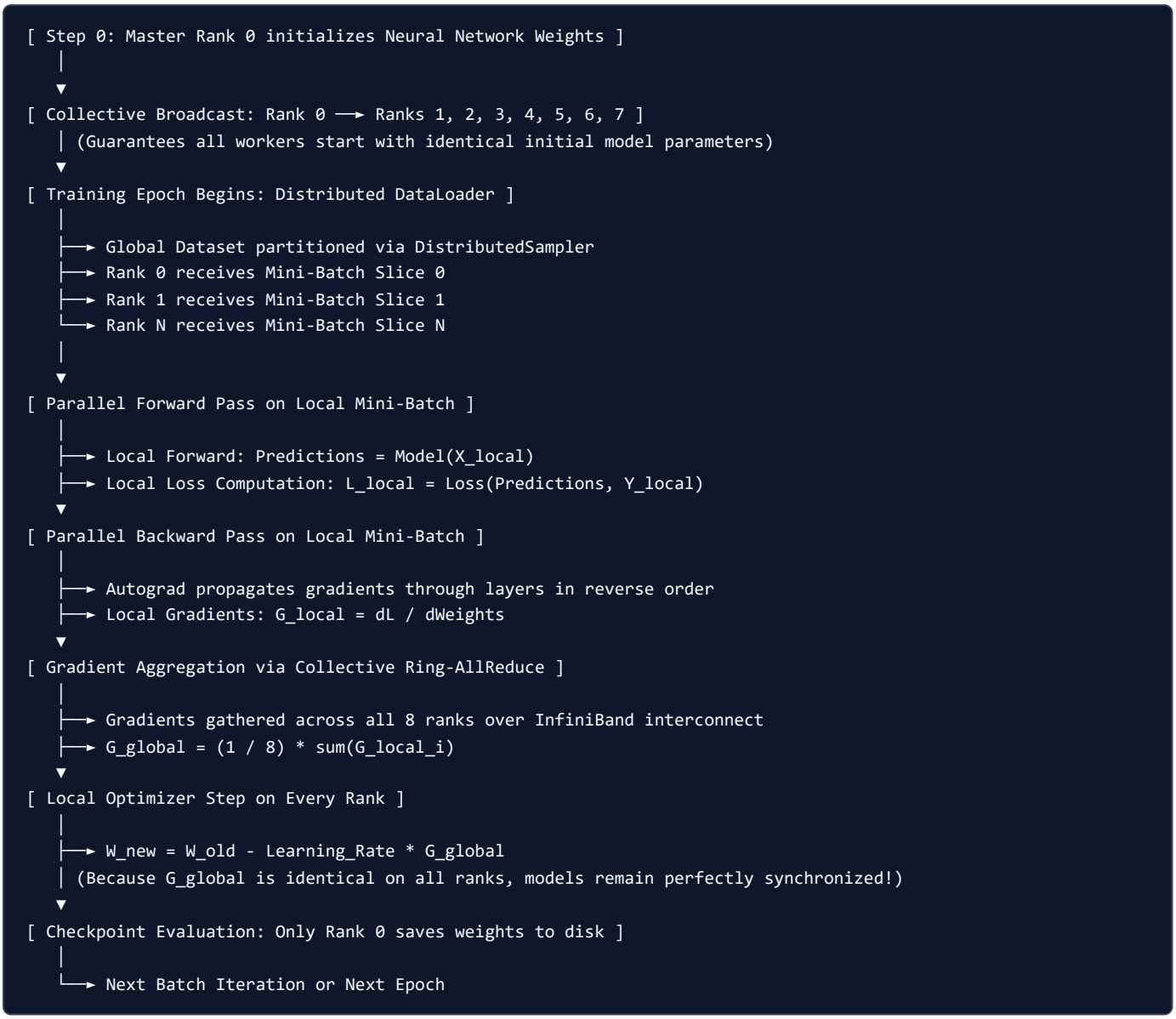
📌 TABLE OF TOPIC TREES

1. Tree 1: Multi-Node Slurm Cluster Job Dispatch Pipeline
2. Tree 2: Distributed Data Parallel (DDP) Gradient Synchronization Loop
3. Tree 3: Horovod Ring-AllReduce Internal Mechanics (Scatter-Reduce & AllGather)
4. Tree 4: Intel VTune Top-down Microarchitecture Analysis Method (TMAM) Decision Tree
5. Tree 5: Intel Advisor Roofline Analysis & Diagnostic Optimization Path
6. Tree 6: InfiniBand RDMA Kernel-Bypass vs Standard TCP/IP Network Flow

TREE 1: Multi-Node Slurm Cluster Job Dispatch Pipeline



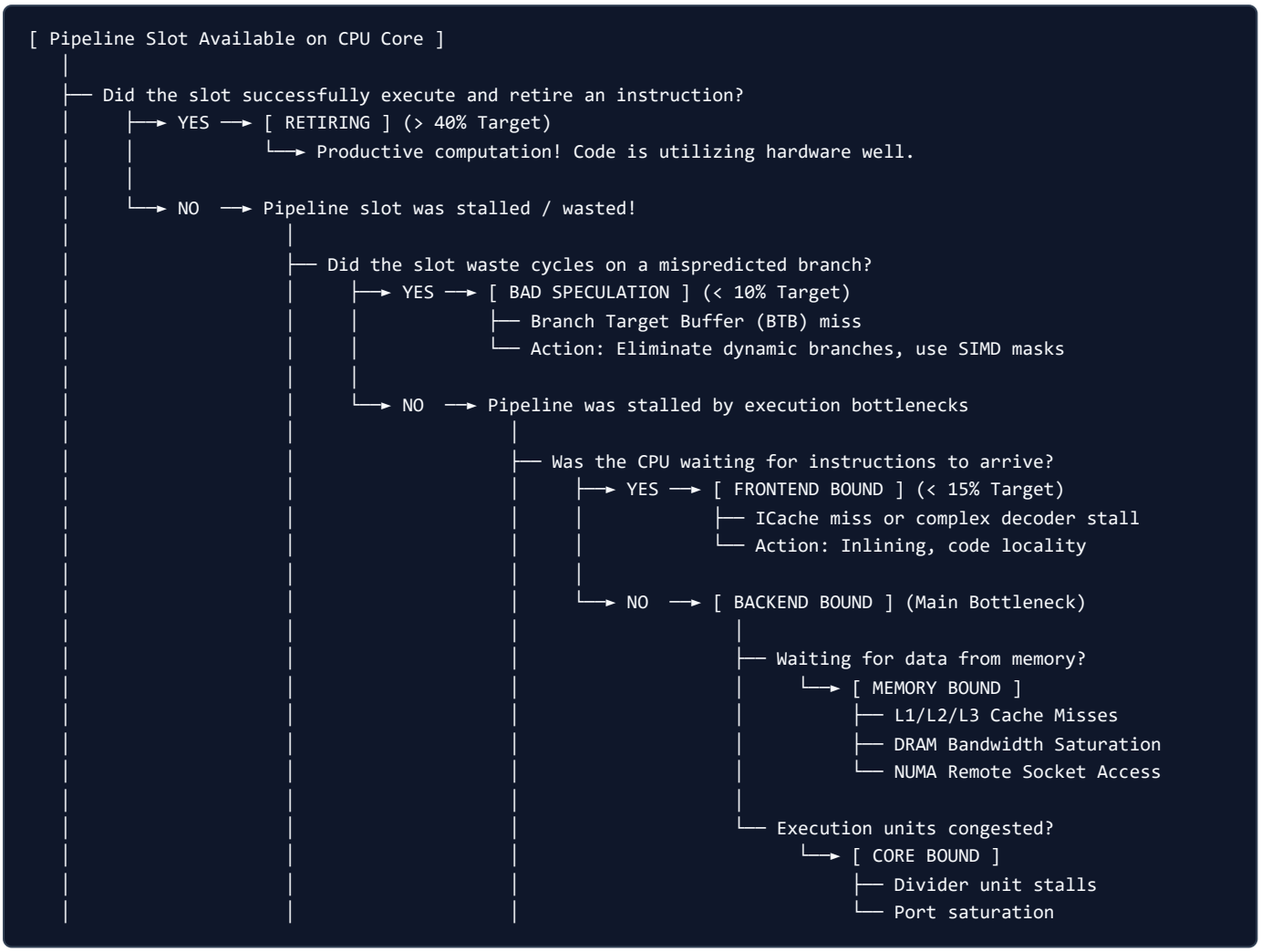
TREE 2: Distributed Data Parallel (DDP) Gradient Synchronization Loop



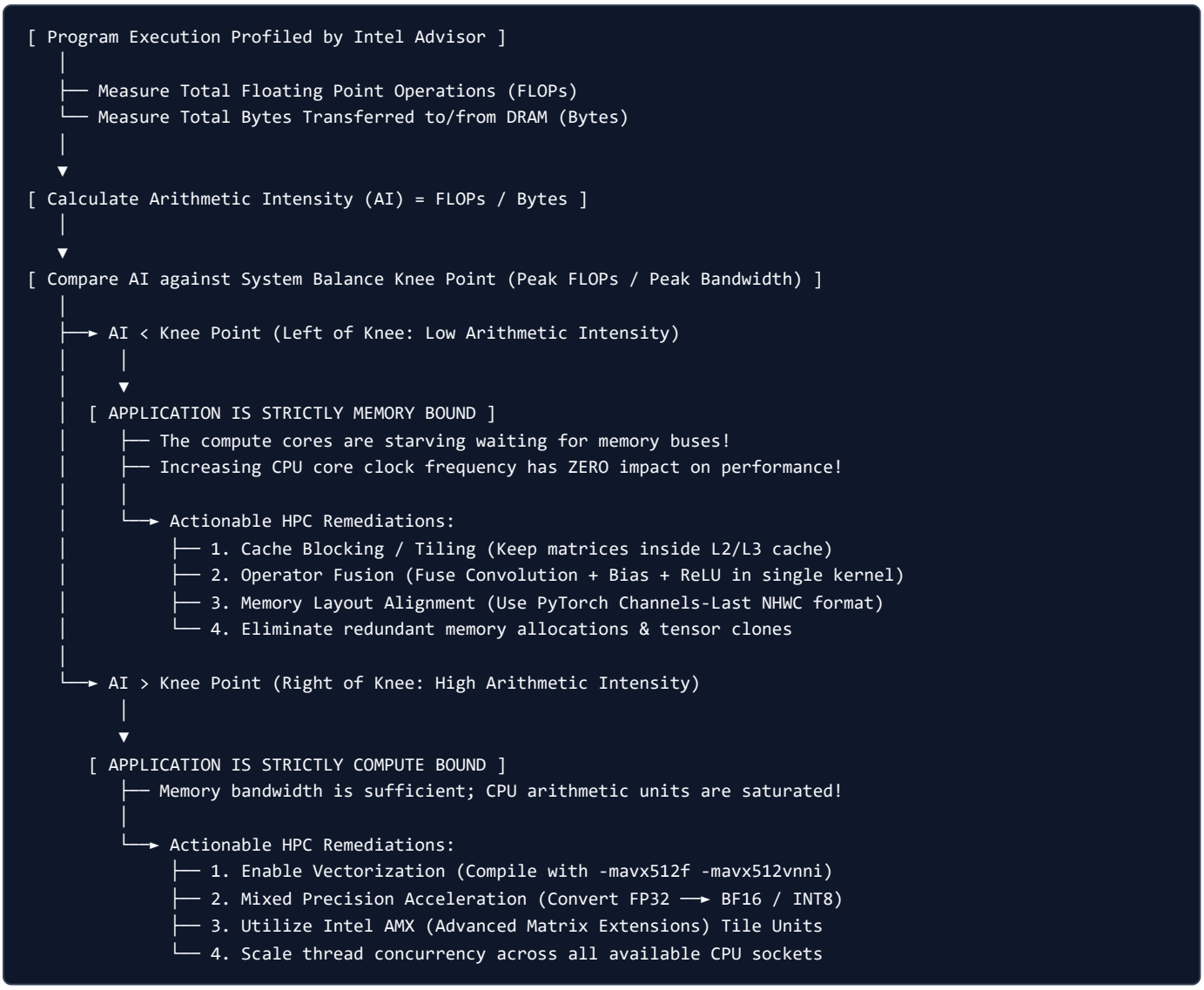
TREE 3: Horovod Ring-AllReduce Internal Mechanics

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[ P = 4 Workers in Logical Ring: Rank 0 ──► Rank 1 ──► Rank 2 ──► Rank 3 ──► Rank 0 ]
   │
   ▼
[ Gradient Tensor Buffer (Size N) divided into P = 4 equal chunks: C0, C1, C2, C3 ]
   │
   ▼
+──────────────────────────────────────────────────────────────────────────────+
| PHASE 1: SCATTER-REDUCE (P - 1 = 3 Transmission Steps)                       |
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   │
   ├──► Step 1:
   │    Rank 0 sends chunk C0 to Rank 1; Rank 1 sums chunk C0 into its buffer
   │    Rank 1 sends chunk C1 to Rank 2; Rank 2 sums chunk C1 into its buffer
   │    Rank 2 sends chunk C2 to Rank 3; Rank 3 sums chunk C2 into its buffer
   │    Rank 3 sends chunk C3 to Rank 0; Rank 0 sums chunk C3 into its buffer
   │
   ├──► Step 2:
   │    Each rank sends the updated chunk to successor for next chunk reduction
   │
   ├──► Step 3:
   │    Final Scatter-Reduce step completes!
   │    Result: Each of the 4 ranks now holds ONE fully reduced chunk:
   │    - Rank 0 holds Sum(C3)
   │    - Rank 1 holds Sum(C0)
   │    - Rank 2 holds Sum(C1)
   │    - Rank 3 holds Sum(C2)
   │
   ▼
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| PHASE 2: ALLGATHER (P - 1 = 3 Transmission Steps)                            |
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   │
   ├──► Step 1:
   │    Each rank sends its fully reduced chunk to its neighbor in the ring
   │    Rank 0 sends Sum(C3) to Rank 1
   │    Rank 1 sends Sum(C0) to Rank 2
   │    Rank 2 sends Sum(C1) to Rank 3
   │    Rank 3 sends Sum(C2) to Rank 0
   │
   ├──► Step 2 & 3:
   │    Transmission continues until all chunks circulate the full ring!
   │
   ▼
[ Complete! Every rank now holds all P fully reduced chunks: Sum(C0, C1, C2, C3) ]
[ Total Data Moved per Rank = 2 * ((P - 1) / P) * N (Independent of Cluster Size!) ]
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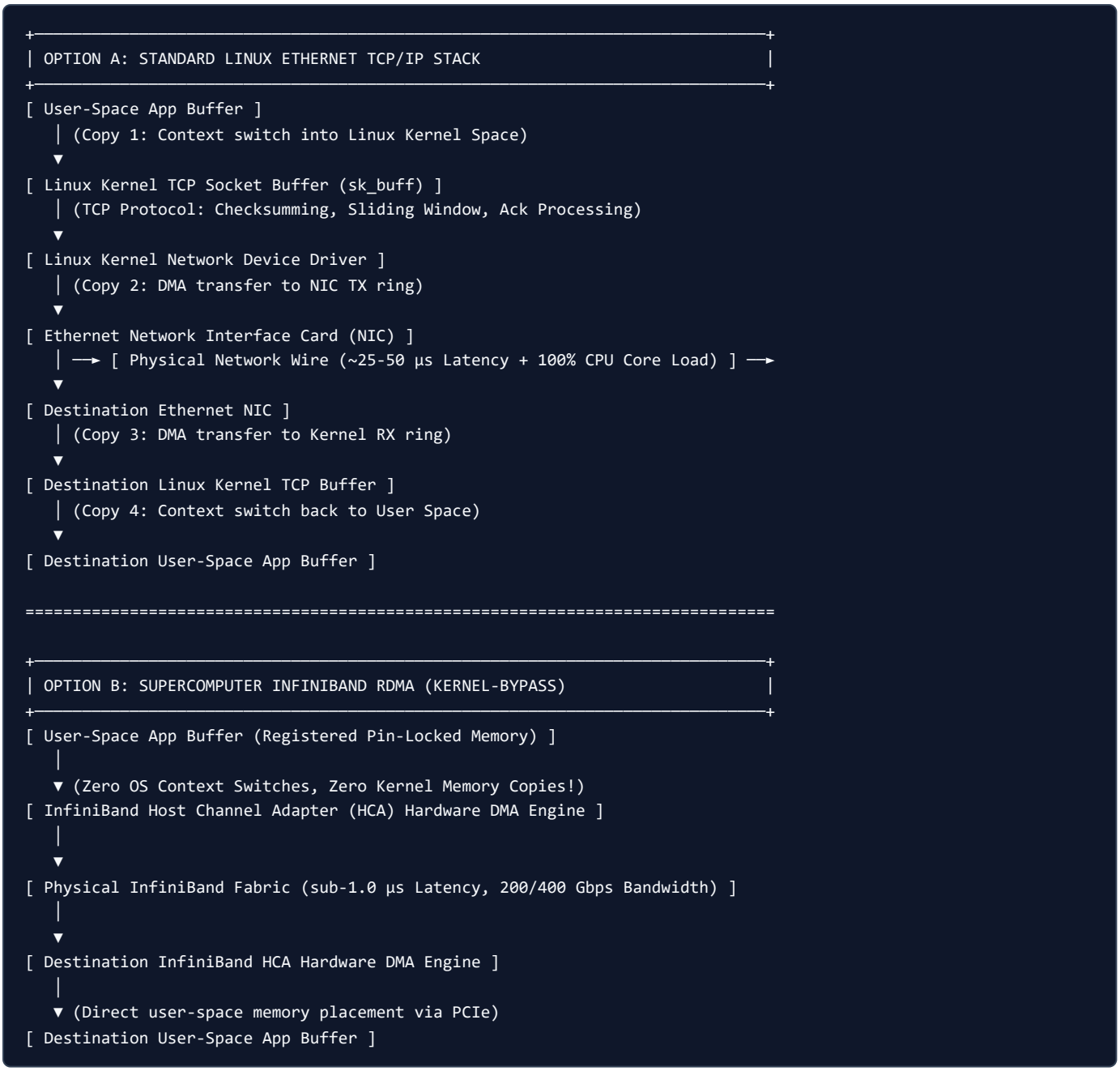
TREE 4: Intel VTune TMAM Decision Tree



TREE 5: Intel Advisor Roofline Diagnostic Path



TREE 6: InfiniBand RDMA vs Standard TCP/IP Flow



GLA University HPC Masterclass | Day 4 Arrow Workflows & Topic Trees Completed